Airline Passenger Satisfaction Prediction Using Deep Learning

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1. Problem Definition

1.1 Introduction

Airline passenger satisfaction is an important factor in the aviation industry. Understanding passenger feedback helps airlines identify areas that require improvement and enhance the overall travel experience.

This project uses machine learning and deep learning techniques to predict whether an airline passenger is satisfied or neutral/dissatisfied based on demographic information, travel details, and service ratings.

1.2 Problem Statement

The objective is to develop a classification model that predicts passenger satisfaction using historical airline passenger data. The project compares traditional machine learning algorithms with Artificial Neural Networks (ANN) to identify a suitable predictive model.

1.3 Objectives

- Analyze airline passenger data to identify patterns and trends.
- Perform data cleaning and exploratory data analysis.
- Create meaningful features from existing service ratings.
- Encode categorical variables and scale numerical features.
- Select relevant features using correlation analysis.
- Train and compare multiple classification models.
- Evaluate the final model using classification metrics.
- Prepare the model for deployment through a FastAPI application.

1.4 Type of Problem

This is a **supervised binary classification problem** because the dataset contains a known target variable, Satisfaction, with two encoded classes.

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2. Dataset Description

2.1 Dataset Source

The dataset was obtained from Maven Analytics.

Dataset link: https://mavenanalytics.io/data-playground/airline-passenger-satisfaction

2.2 Dataset Overview

- **Total records:** 129,880
- **Original columns:** 24
- **Target variable:** Satisfaction
- **Problem type:** Binary classification
- **Identifier handling:** The passenger ID column was removed because it is an identifier rather than a meaningful predictive feature.
- **After Handling identifier and target column:** 22

2.3 Feature Categories

The dataset contains the following types of information:

- **Demographic information:** Gender, Age and Customer Type.
- **Travel information:** Type of Travel, Class and Flight Distance.
- **Delay information:** Departure Delay and Arrival Delay.
- **Service ratings:** Seat Comfort, Online Boarding, In-flight Entertainment, Baggage Handling and other passenger service ratings.
- **Target variable:** Satisfaction.

2.4 Target Variable

The target variable is Satisfaction, which contains two categories:

- Satisfied
- Neutral or Dissatisfied

The target was encoded into numerical values for model training.

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3. Exploratory Data Analysis (EDA)

3.1 Dataset Inspection

The initial dataset was inspected using methods such as head(), shape, info() and describe() to understand its structure, column types and statistical characteristics.

This step helped identify numerical variables, categorical variables and potential data quality issues.

3.2 Missing Value Analysis

The initial dataset contained **393 missing values** in the arrival-delay column, representing approximately **0.30% of the records**. No other missing values were reported in the initial dataset.

The missing arrival-delay values were handled during data cleaning to ensure that the model could train on valid input data.

3.3 Duplicate Analysis

The initial dataset contained **zero duplicate records**. Therefore, no duplicate rows needed to be removed.

3.4 Target Distribution

The target distribution was analyzed to understand the proportion of satisfied and neutral/dissatisfied passengers.

Satisfaction category	Number of passengers	Percentage
Neutral or Dissatisfied	73,452	56.55%
Satisfied	56,428	43.45%
Total	**129,880**	**100%**

The classes are moderately imbalanced, with neutral/dissatisfied passengers forming the larger group. Therefore, precision, recall and F1-score were considered alongside accuracy when evaluating the models.

3.5 Univariate Analysis

Univariate analysis was performed to understand the distribution of individual variables.

Numerical distributions and categorical frequency counts were examined to identify common passenger characteristics, travel patterns and service-rating distributions.

3.6 Bivariate Analysis

Bivariate analysis was used to examine relationships between selected features and passenger satisfaction.

The analysis helped explore whether demographic characteristics, travel class, flight delays and service ratings were associated with different satisfaction outcomes.

3.7 Correlation Analysis

A correlation matrix was used to examine the linear relationships between numerical features.

The analysis helped identify features with stronger positive or negative correlations with satisfaction and highlighted relationships between certain input variables.

Correlation indicates association rather than causation.

3.8 Outlier Analysis

Potential outliers were examined in numerical columns, including age, flight distance, departure delay and arrival delay.

Extreme delay values may represent genuine flight disruptions rather than data-entry errors. Therefore, observations were not automatically removed solely because they were extreme. Retaining them allows the model to learn from a wider range of passenger experiences.

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4. Data Cleaning and Preprocessing

4.1 Removing Unnecessary Columns

The passenger ID column was removed because it uniquely identifies records and does not directly describe passenger satisfaction.

4.2 Handling Missing Values

The missing values in the arrival-delay column were handled before model training. This ensured that the processed dataset did not contain missing values in the model inputs.

4.3 Encoding Categorical Variables

Categorical variables were converted into numerical representations.

- **Label Encoding:** Applied to Satisfaction, Gender, Customer Type and Type of Travel.
- **One-Hot Encoding:** Applied to Class to create separate binary columns representing travel classes.

These transformations made the categorical information suitable for use by the selected algorithms.

4.4 Separating Features and Target

The independent variables were separated into `X`, while the target variable `Satisfaction` was stored in `y`.

This separation allowed the models to learn the relationship between passenger attributes and satisfaction outcomes.

4.5 Train, Validation and Test Split

The dataset was divided into training, validation and test sets.

- **Training set:** 70 % Used to fit the models.
- **Validation set:** 15% Used to compare models and monitor performance during model development.
- **Test set:** 15% Used for the final evaluation of the selected model.

Keeping the test set separate helps provide a more reliable estimate of performance on unseen data.

4.6 Feature Scaling

`StandardScaler` was used to standardize the selected numerical input features.

The scaler was fitted on the training data only. The same fitted scaler was then applied to the validation and test data to prevent information from these sets influencing the scaling parameters.

Scaling is especially useful for neural networks because it places numerical features on more comparable scales.

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5. Feature Engineering

5.1 Objective

Feature engineering was performed to create meaningful summary variables from related passenger service ratings.

The goal was to represent broader aspects of the passenger experience using combinations of existing features.

5.2 Digital Experience Score

The Digital Experience Score was calculated using the mean of the following ratings:

- Ease of Online Booking
- Online Boarding
- In-flight Wifi Service

This feature represents the passenger's experience with digital and online airline services.

5.3 Comfort Score

The Comfort Score was calculated using the mean of:

- Seat Comfort
- Leg Room Service
- On-board Service

This feature summarizes selected aspects of passenger comfort during the journey.

5.4 Service Experience Score

The Service Experience Score was calculated using the mean of:

- Departure and Arrival Time Convenience

- Check-in Service
- Gate Location
- In-flight Service
- In-flight Entertainment
- Baggage Handling
- Cleanliness
- Food and Drink

This feature provides an aggregate representation of the passenger's experience across selected airline services.

5.5 Benefits of Feature Engineering

- Combines related ratings into interpretable summary features.
- Helps represent broader aspects of the passenger experience.
- May reduce the need for a model to learn every relationship independently.
- Provides additional information for evaluating passenger satisfaction.

These engineered features were evaluated as part of the model-development process.

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6. Feature Selection

6.1 Objective

Feature selection was performed to identify the 15 input features with the strongest absolute linear correlations with Satisfaction.

6.2 Correlation-Based Method

Pearson correlation was calculated between each input feature and the target variable using the training data.

Absolute correlation values were used to rank features because a strong negative correlation can be just as informative as a strong positive correlation.

The 15 features with the largest absolute correlation values were selected.

6.3 Feature Selection Procedure

1. Calculate feature-to-target correlations using training data only.
2. Take the absolute values of the correlations for ranking.
3. Select the 15 features with the highest absolute correlation.
4. Display their original signed correlation values.
5. Visualize the selected features using a bar plot.
6. Apply the same selected feature names and order to the training, validation and test sets.

6.4 Advantages

- Reduces the number of model input features.
- Provides an interpretable feature-ranking method.
- Helps identify features with strong linear associations with satisfaction.
- Can reduce model complexity and unnecessary inputs.

6.5 Limitations

Correlation-based selection may overlook features that have weak individual linear correlations but useful nonlinear relationships or interactions with other variables.

It also does not establish that a feature causes satisfaction.

6.6 Conclusion

The top 15 features were selected based on their absolute correlation with the target. The usefulness of this reduced feature set was assessed through subsequent model training and evaluation.

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7. Model Development

7.1 Models Evaluated

The project evaluated traditional machine learning models and neural network models to compare their predictive performance.

The models included:

1. Logistic Regression
2. Random Forest Classifier
3. Baseline Artificial Neural Network
4. Improved Artificial Neural Network
5. Final Artificial Neural Network

7.2 Logistic Regression

Logistic Regression was used as a baseline classification model.

It estimates the probability of a passenger belonging to a particular satisfaction class. Its performance provides a reference point for evaluating more complex models.

7.3 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees.

It can capture nonlinear relationships and interactions between input features. It was included to compare a tree-based model against the neural network approaches.

7.4 Baseline Artificial Neural Network

A baseline ANN was trained to learn patterns between passenger information and satisfaction.

The network uses dense layers to learn feature relationships and a sigmoid output for binary classification.

This model established a baseline for evaluating improvements to the neural network architecture and training process.

7.5 Improved Artificial Neural Network

The improved ANN introduced architectural and training adjustments, including Batch Normalization, Dropout and training callbacks.

- **Batch Normalization:** Helps stabilize the distribution of activations during training.
- **Dropout:** Randomly disables selected neurons during training to help reduce overfitting.
- **Early Stopping:** Stops training when validation loss stops improving and restores the best weights.
- **ReduceLROnPlateau:** Reduces the learning rate when validation loss stops improving.

These techniques were used to improve training stability and encourage generalization.

7.6 Final Artificial Neural Network

The final ANN used the following architecture in the documented experiment:

- Input layer matching the selected feature count.
- Dense layer with 256 neurons and ReLU activation.
- Batch Normalization.
- Dropout with a rate of 0.4.
- Dense layer with 64 neurons and ReLU activation.
- Batch Normalization.
- Dropout with a rate of 0.4.
- Output layer with one neuron and sigmoid activation.

Training configuration:

- Optimizer: Adam
- Learning rate: 0.0005
- Loss function: Binary Cross-Entropy
- Maximum epochs: 50
- Batch size: 32
- Early Stopping patience: 8
- ReduceLROnPlateau factor: 0.5
- ReduceLROnPlateau patience: 3
- Minimum learning rate: 0.00001

The configuration was designed to balance learning capacity with regularization. The final choice was evaluated using held-out test data.

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8. Model Evaluation

8.1 Evaluation Metrics

The following metrics were used to assess classification performance.

Accuracy: Measures the proportion of all predictions that are correct.

Precision: Measures the proportion of positive predictions that are correct.

Recall: Measures the proportion of actual positive cases that the model correctly identifies.

F1-score: Combines precision and recall using their harmonic mean.

Loss: Measures the prediction error according to the selected loss function. Lower loss generally indicates better predictions when comparing the same loss definition and evaluation data.

8.2 Validation Model Comparison

The following validation results were reported in one model-comparison run:

Model	Accuracy	F1-score
Logistic Regression	86.01%	83.70%
Random Forest	94.90%	94.04%
Baseline ANN	95.33%	94.54%
Improved ANN	95.23%	94.43%

The results indicate that the ANN models and Random Forest performed substantially better than Logistic Regression in this comparison. The baseline ANN achieved the highest reported validation accuracy and F1-score among these four models.

The final ANN was evaluated separately on the test set. Validation and test results should not be mixed when making direct performance comparisons.

8.3 Final ANN Test Results

The latest reported final ANN test evaluation produced the following metrics:

Metric	Score
Test Loss	0.1154
Accuracy	95.27%
Precision	95.90%
Recall	93.10%
F1-score	94.48%

These results indicate that the final ANN correctly classified approximately 95 out of every 100 passengers in the test set.

The precision score indicates that most passengers predicted as satisfied were actually satisfied. The recall score indicates that the model identified approximately 93% of the passengers belonging to the encoded positive class.

The interpretation of precision and recall depends on the target encoding. In this project, class 1 represents Satisfied.

8.4 Model Selection Conclusion

The ANN models demonstrated strong predictive performance in the reported experiments. The final ANN was evaluated on the held-out test set and achieved an accuracy of 95.27% and an F1-score of 94.48%.

The final model provides a promising balance of predictive performance and nonlinear learning capability. The model

selection should also consider the validation results, generalization performance, deployment requirements and the intended use of the predictions.

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9. Confusion Matrix Analysis

9.1 Purpose

A confusion matrix provides a detailed view of correct and incorrect classifications for each target class.

For this binary classification problem, the matrix contains:

- **True Negative (TN):** Neutral/dissatisfied passengers correctly predicted as neutral/dissatisfied.
- **False Positive (FP):** Neutral/dissatisfied passengers incorrectly predicted as satisfied.
- **False Negative (FN):** Satisfied passengers incorrectly predicted as neutral/dissatisfied.
- **True Positive (TP):** Satisfied passengers correctly predicted as satisfied.

	Predicted Neutral/Dissatisfied	Predicted Satisfied
Actual Neutral/Dissatisfied	10,681 (TN)	337 (FP)
Actual Satisfied	584 (FN)	7,880 (TP)

9.2 Interpretation

The confusion matrix helps identify which type of classification error occurs more frequently.

False negatives represent satisfied passengers who were not recognized as satisfied. False positives represent passengers predicted to be satisfied even though their actual class was neutral/dissatisfied.

The exact counts should be taken from the confusion matrix generated using the final model's test predictions.

9.3 Importance

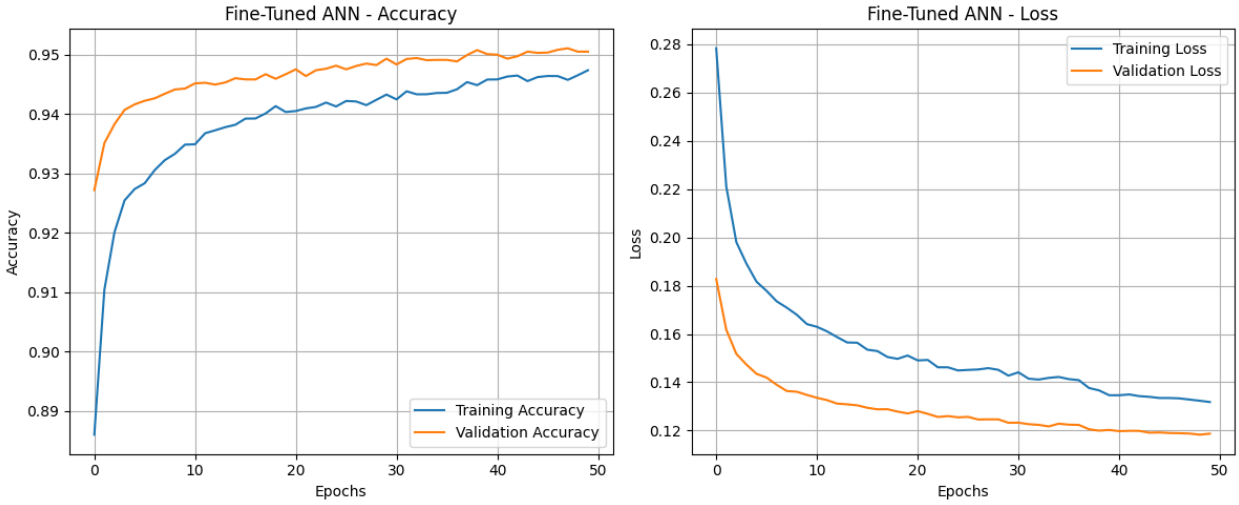
This analysis complements accuracy by showing the distribution of errors across the two classes. It helps determine whether further model adjustments are needed to improve the identification of satisfied or dissatisfied passengers.

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10. Overfitting and Underfitting Analysis

10.1 Overfitting

Overfitting occurs when a model learns the training data too closely and performs less effectively on unseen data.



10.2 Underfitting

Underfitting occurs when a model cannot learn the patterns in the data sufficiently well.

OBSERVED no sign of underfitting and overfitting in the model.

10.3 Techniques Used to Reduce Overfitting

The final ANN used Batch Normalization, Dropout, Early Stopping and learning-rate reduction.

These techniques can improve training stability and reduce overfitting risk. However, whether overfitting was actually reduced must be determined by examining the training and validation curves.

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11. Error Analysis

11.1 Objective

Error analysis was performed to understand the types of predictions made incorrectly by the final ANN.

11.2 Major Errors

The main error categories are false positives and false negatives.

- **False positives:** Passengers predicted as satisfied when their actual class was neutral/dissatisfied.
- **False negatives:** Satisfied passengers incorrectly predicted as neutral/dissatisfied.

The relative frequency of these errors can be obtained from the final test confusion matrix.

11.3 Possible Reasons for Misclassification

Potential reasons include:

- Different passengers may give similar service ratings but have different overall satisfaction.
- Several service-related features may contain overlapping information.
- Some satisfaction outcomes may depend on factors not represented in the dataset.
- Borderline passenger experiences may be difficult to distinguish.
- The model may not capture every useful nonlinear relationship.

These are possible explanations, not confirmed causes. They should be investigated by examining misclassified records and their feature values.

11.4 Possible Improvements

- Inspect false-positive and false-negative samples.
- Compare training and validation curves.
- Review the selected feature set.
- Tune network architecture and regularization.
- Evaluate class-specific precision and recall.
- Check that all preprocessing steps are consistent between training and inference.

11.5 Conclusion

The final ANN achieved strong overall test performance, but some classification errors remain. Error analysis provides a practical way to identify areas for further improvement.

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12. Data Leakage and Correlated Features

12.1 Data Leakage Prevention

Data leakage occurs when information unavailable at prediction time, or information derived from the evaluation data, influences model training.

The workflow uses training data to calculate feature correlations and fit the scaler. The same selected features and fitted scaler are then applied to validation and test data.

The preprocessing and feature-selection steps should be reproduced consistently during deployment.

12.2 Correlated Features

Some airline ratings may be strongly correlated because they measure related aspects of the passenger experience.

Highly correlated inputs do not automatically reduce ANN accuracy. However, they may introduce redundant information, affect model interpretation or provide limited additional predictive value.

12.3 Important Practical Consideration

Because the target is passenger satisfaction and many inputs are service ratings, it is important to consider when those ratings become available.

If the model is intended to predict satisfaction before a flight, ratings collected after the flight would not be available and could make the prediction unrealistic. The feature set must match the intended prediction time.

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13. Model Saving and Deployment

13.1 Objective

The deployment stage prepares the trained ANN for use outside the training notebook.

A FastAPI application can accept passenger details, apply the same preprocessing used during training and return a predicted satisfaction class.

13.2 Model and Preprocessing Artifacts

The deployment workflow may require the following files:

- ann_model.keras — trained ANN model.
- scaler.pkl — fitted StandardScaler.
- selected_features.pkl — selected feature names in the correct order.
- label_encoders.joblib — categorical encoding mappings, if required by the application.
- class_columns.pkl — one-hot encoded Class column names, if used by the application.

Only the artifacts actually generated by the notebook and required by the inference code should be included.

13.3 FastAPI Application

FastAPI can expose an API endpoint that accepts passenger information and returns the predicted satisfaction category.

The application must reproduce the training preprocessing, including categorical encoding, feature engineering, feature ordering and scaling. The input schema must match the features required to calculate the model inputs.

13.4 Deployment Validation

Before deployment, the following should be checked:

- The saved model loads successfully.
- The preprocessing artifacts load successfully.
- The input features match the expected names and order.
- Missing or invalid inputs are handled appropriately.
- Predictions are returned in the correct class format.
- Predictions from the API are consistent with predictions made in the notebook for the same input.

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14. Limitations

The project has several limitations:

- Correlation-based feature selection focuses on individual linear relationships and may miss useful nonlinear interactions.
- Passenger ratings are subjective and may not capture every factor affecting satisfaction.
- Model performance depends on the quality and representativeness of the dataset.
- High test accuracy does not guarantee identical performance on future passengers or different airlines.
- Deployment predictions are reliable only when the input features and preprocessing match the training workflow.
- If service ratings are unavailable at the intended prediction time, the model would require a revised feature set.

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15. Future Enhancements

Possible future improvements include:

- Comparing correlation-based feature selection with model-based methods.
- Evaluating feature importance using suitable explainability techniques.
- Tuning hyperparameters using a controlled validation procedure.
- Examining class-specific errors and calibration.
- Testing the model on additional or more recent airline passenger data.
- Building a user-friendly interface around the FastAPI endpoint.
- Monitoring prediction quality after deployment.

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16. Final Conclusion

This project developed an airline passenger satisfaction classification workflow using exploratory data analysis, data preprocessing, feature engineering, feature selection and model evaluation.

Logistic Regression, Random Forest and several ANN configurations were evaluated. In the reported test evaluation, the final ANN achieved **95.27% accuracy, 95.90% precision, 93.10% recall and a 94.48% F1-score**.

The results demonstrate that deep learning can be effective for identifying patterns in passenger satisfaction data. The project also highlights the importance of consistent preprocessing, careful feature selection, appropriate evaluation metrics and error analysis.

The completed workflow provides a foundation for further experimentation and potential API-based deployment. Any real-

world use should also ensure that the chosen input features are available at the time predictions are required.

